

OViP: Online Vision-Language Preference Learning

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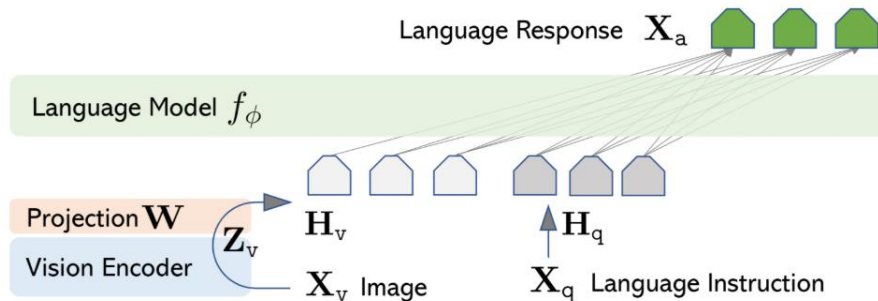
04 | Conclusion

01

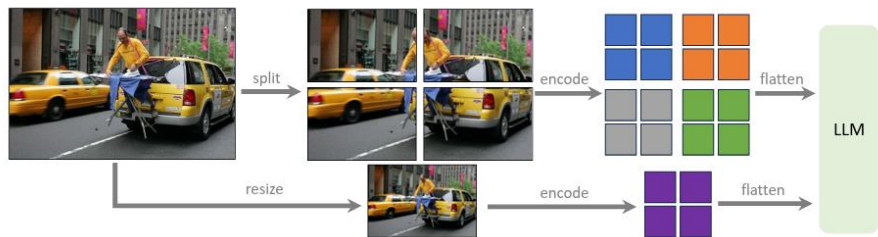
Introduction

01. Introduction

1. LVLMs (Large Vision-Language Models)



LLaVA-1.0



LLaVA-1.5

Visual input example, Chicken Nugget Map:

Sometimes I just look at pictures of the earth from space and I marvel at how beautiful it all is.



User
LLaVA

Can you explain this meme in detail?

The meme in the image is a creative and humorous take on food, with a focus on chicken nuggets, as the center of the universe. The meme begins with the phrase "Sometimes I just look at pictures of the earth from space and I marvel at how beautiful it all is..." The main image features a tray of cooked chicken nuggets, with the chicken pieces positioned to resemble the continents and islands on a world map. The meme playfully suggests that the chicken nuggets represent the Earth, and the various locations depicted in the photo are actually chicken nugget versions of different places. The punchline of the meme appears in smaller text at the bottom, reading "I mean, it's not the real Earth, but how beautiful it is all is." This lighthearted and entertaining meme combines food and humor, using the image of chicken nuggets to create a funny and imaginative representation of the world.

User
GPT-4 [32]

Can you explain this meme?

This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets. The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world. The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.

User
BLIP-2

Can you explain this meme in detail?

sometimes i just look at pictures of the earth from space and marvel how beautiful it is

User

Can you explain this meme in detail?

OpenFlamingo

It's a picture of a chicken nugget on the International Space Station.

01. Introduction

1. LVLMs (Large Vision-Language Models)

» Hallucination Problem



Describe the image in detail.



Regular LLaVA 1.5

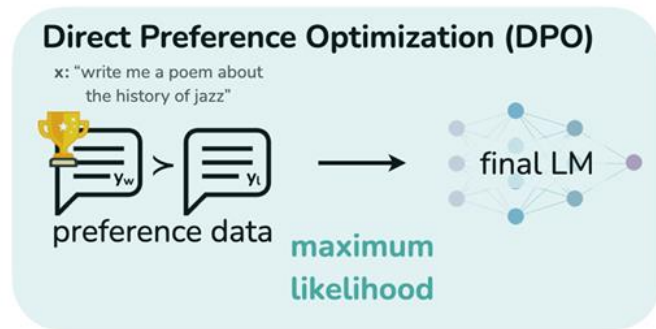
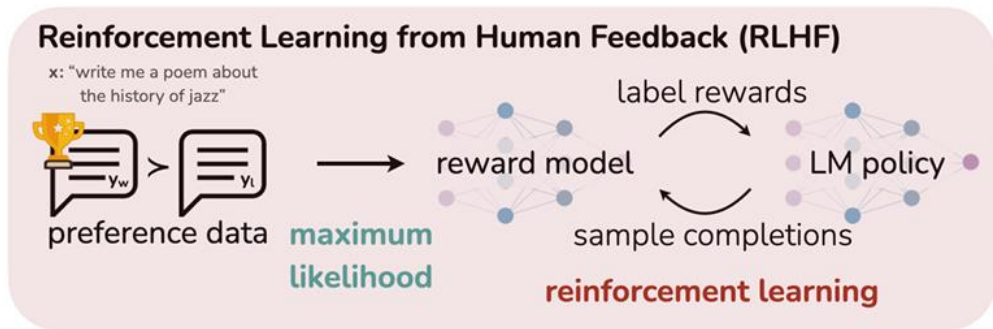
The image depicts a **rainy day** in a city, with **wet sidewalks** reflecting the surrounding environment. There are several people walking along the sidewalks, carrying umbrellas to shield themselves from the rain. The street is filled with various vehicles, including cars, **buses, and trucks**. **A bus can be seen in the middle of the scene.**

01. Introduction

2. DPO (Direct Preference Optimization)

» Direct Preference Optimization: Your Language Model is Secretly a Reward Model

- 딥러닝논문읽기모임(2024년 5월 8일), 펀디멘탈팀 차지윤님 발표 참고
 - [Direct Preference Optimization Your Language Model is Secretly a Reward Model - YouTube](#)
- DPO의 성공이 Hallucination 완화를 위한 Multi-modal DPO 연구 촉진



$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\underbrace{\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)}}_{\text{선호 데이터 Probability}} - \underbrace{\beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)}}_{\text{비선호 데이터 Probability}} \right) \right]$$

01. Introduction

3. OViP (Online Vision-Language Preference Learning)

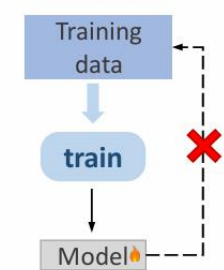
» 기존 방식

- **Textual Response**만 사용하여 LLM용 DPO를 확장하여 LVLM에 적용
 - Preference Optimization에만 집중
- 동일한 **Text Response** 기반 **Preference Pair** 구성하여 **Visual, Text Preference Optimization** 수행
 - Visual Content에 집중하지만, 모델의 Hallucination 결과와 명시적 관련 없음

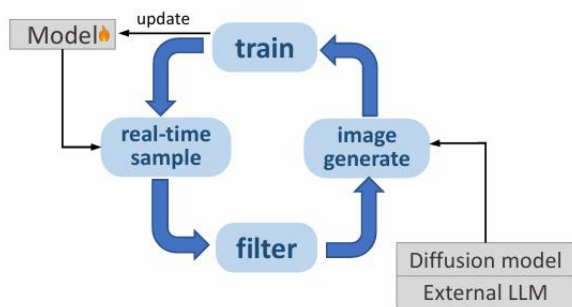
01. Introduction

3. OViP (Online Vision-Language Preference Learning)

OViP = Online Preference Learning + Image-aware



(a) offline Training



(b) OViP Training

Q: How many bats are pictured?



There are **no** bats. (green box)
There are **two** bats. (orange box)

$P_{\pi}(\text{two}|Q, \text{Image}; \text{There are } \text{)} = ?$
Base: 70.9%; **Offline: 42.7%**; **Online: 7.1%**

$P_{\pi}(\text{no}|Q, \text{Image}; \text{There are } \text{)} = ?$
Base: 2.5%; **Offline: 39.5%**; **Online: 88.2%**

Online Preference learning
More valid optimization

Q: Why is the cow in the truck?



A: The cow is not in the truck. It appears to be standing by a fence

After Training

$\log P_{\pi}(A|Q, \text{Image}) = ?$

- Base: -64.9
- **w/o Image: -63.1** (overfitting!)
- **w. Image: -71.1**

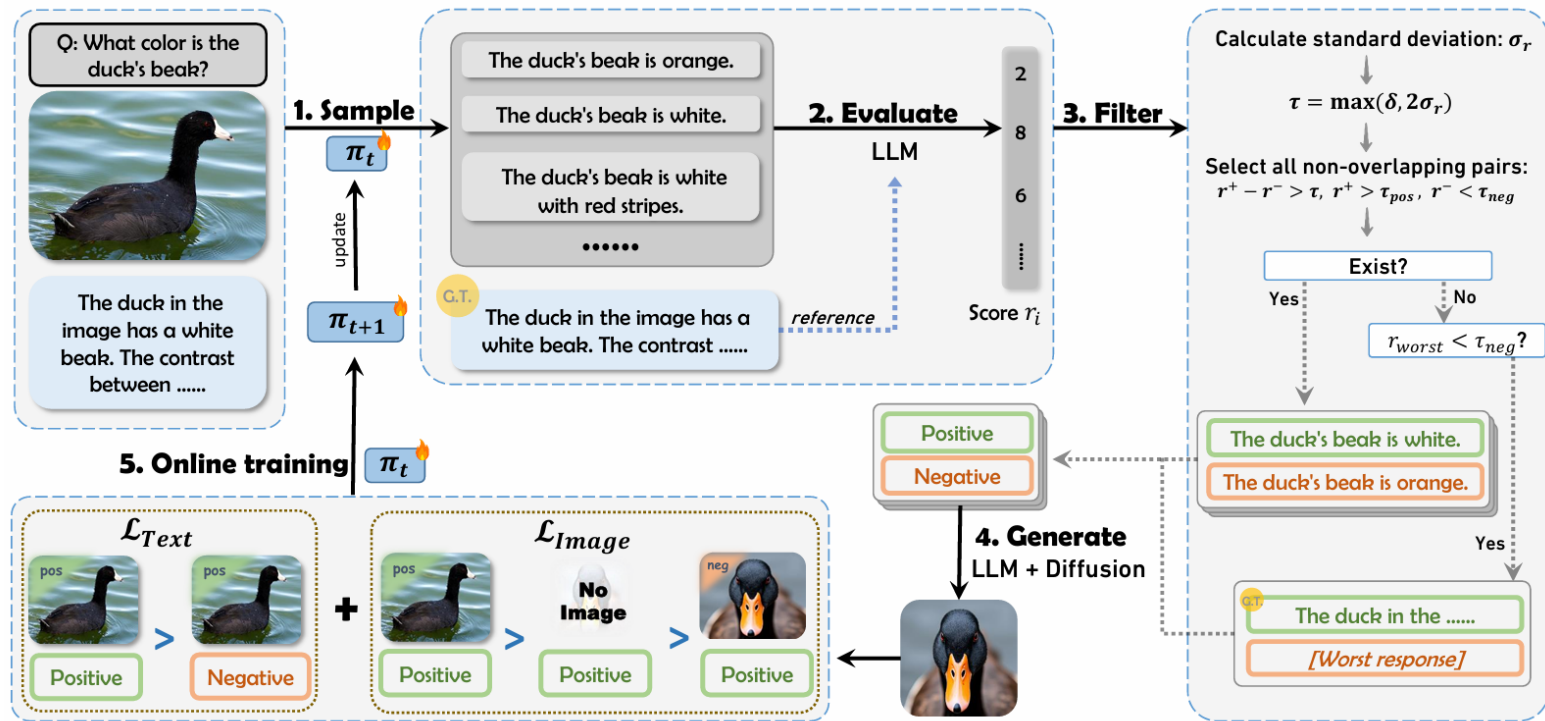
Image-aware training
Less overfitting

02

Method

02. Method

1. Overview



02. Method

Sample

Evaluate

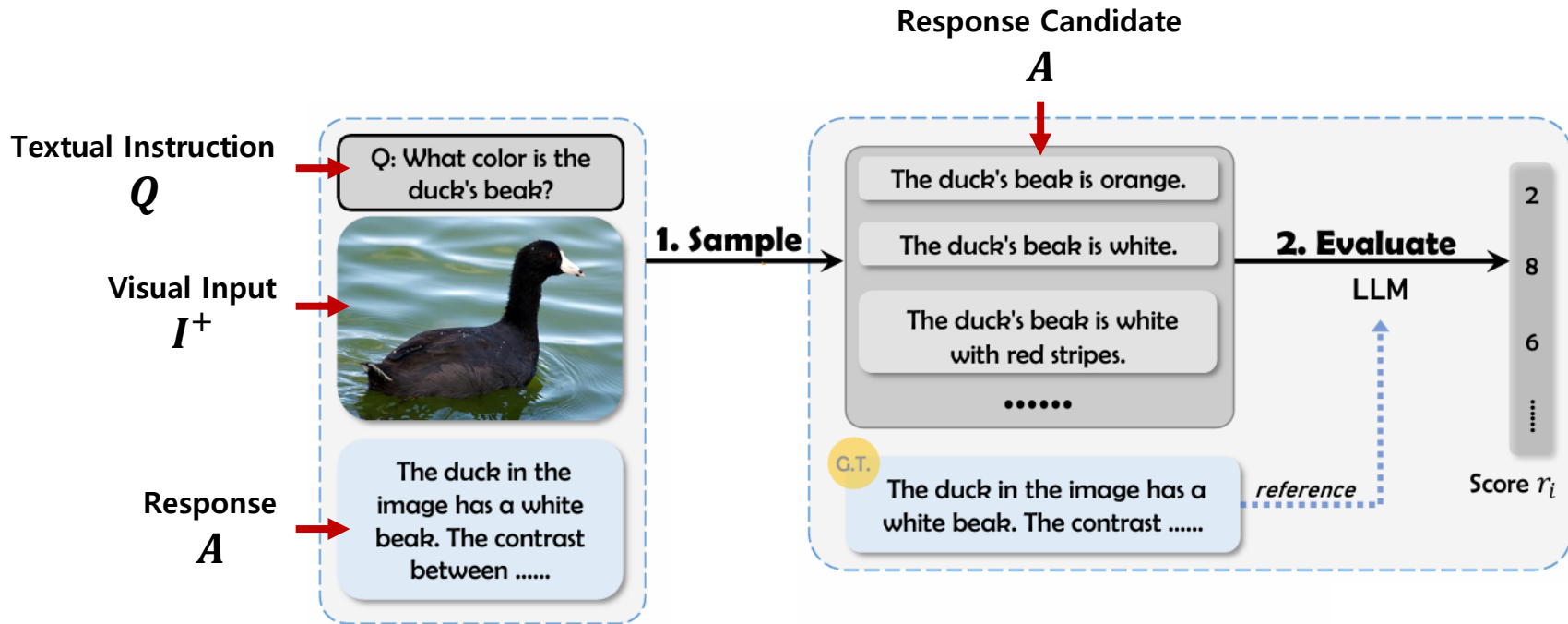
Filter

Generate

Online
Training

2. Real-time Generation of Output Data

» Stochastic Sampling($K=16$)을 통해 Response Candidate 생성



02. Method

Sample

Evaluate

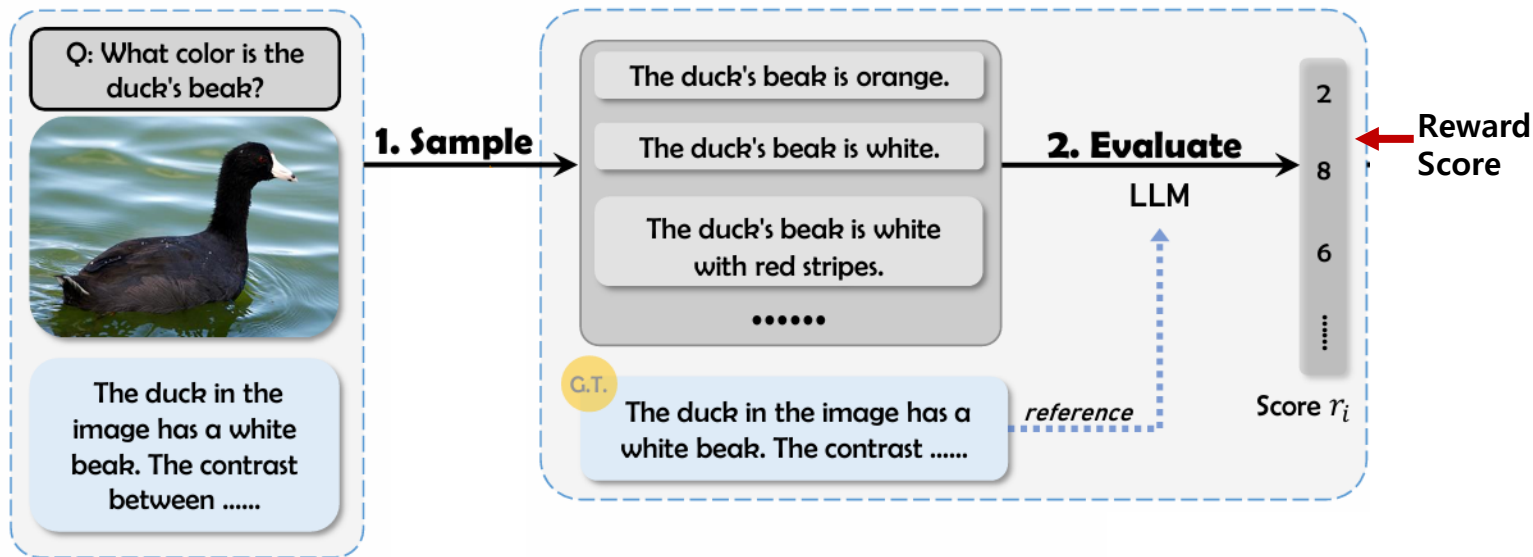
Filter

Generate

Online
Training

2. Real-time Generation of Output Data

» LLM 기반 Reward Function으로 평가, GT Answer와의 일치성을 반영한 Reward Score 계산



02. Method

Sample

Evaluate

Filter

Generate

Online
Training

2. Real-time Generation of Output Data



LLM 기반 Reward Function으로 평가, GT Answer와의 일치성을 반영한 Reward Score 계산

- Reward Model : Qwen-2.5-7b-instrut

Prompt for Quality Judgment

Task

Your role is as a discerning assistant tasked with evaluating model responses for multimodal tasks (though you have no access with the image). Upon being presented with a question that requires the interpretation of both text and images, you will receive two distinct responses. The first is crafted by our sophisticated multimodal model, while the second represents an approximate ideal answer—it may be incomplete. Your objective is to meticulously and precisely assess the model-generated response (the former) based on the provided reference answer (the latter).

- Here's how you should approach the assessment process:

1. The quality of the response depends on its accuracy and the degree of adherence to the correct answer. Therefore, if the response is much more detailed than the reference answer, it should not be considered a very good response (although it may still be considered a good one).
2. Directly provide the score of the response, with a full score of 10. Your response should follow this format: "Score: [x]\n", where "[x]" represents the score you give, and "\n" is a line break.
3. Please do not provide additional reasoning, just give the score directly.

Question

{question}

Response

{response}

Correct answer

{answer}

02. Method

Sample

Evaluate

Filter

Generate

Online Training

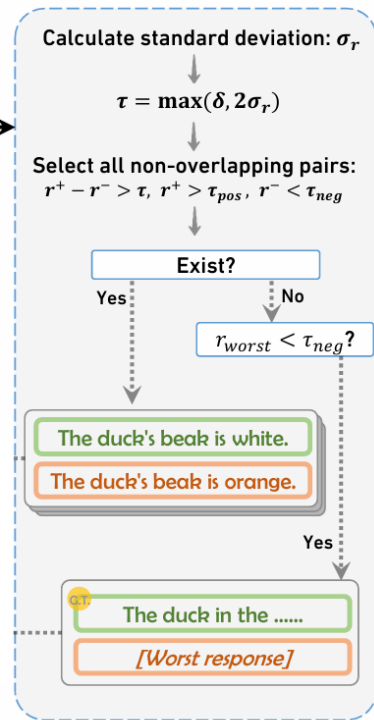
3. Contrasting Response Pair Sampling

» Dynamic Preference Pair : Batch 내에서 Score 차이가 큰 Response Pair 선택

- 품질차이가 큰 Pair 구성하여 효과적으로 학습을 위한 목적
- 미리 정의 된 품질 기준을 정의
- 전반적으로 품질이 저조한 경우 GT를 Positive Response로 설정

$$\mathcal{D}_{\text{pair}} = \left\{ (Q, \mathcal{I}^+, \mathcal{A}^+, \mathcal{A}^-) \mid \mathcal{A}^+, \mathcal{A}^- \in \{\mathcal{A}^i\}_{i=1}^k, \right. \\ \left. |r^+ - r^-| > \max(\delta, 2\sigma_r), \right. \\ \left. r^+ > \tau_{\text{pos}}, r^- < \tau_{\text{neg}} \right\}$$

3. Filter



02. Method

Sample

Evaluate

Filter

Generate

Online Training

4. Inverse Negative Image Synthesis

➤ Negative Image 생성

- 외부 LLM을 사용하여 Response 기반으로 Image 생성 Prompt 생성
 - LLM Model : Qwen-2.5-7b-instruct / Image Generation Model : FLUX.1-dev
- Positive Response와 Negative Response 간 Semantic Difference를 식별하게 하도록 Text Description 생성

Prompt for image generation-Part 1

Task

Given an unknown image-related question, a correct answer, and an inaccurate response, carefully analyze the differences between the response and the answer. Then, provide a brief description of the image so that it aligns with the correct answer and differs from the incorrect responses. In other words, infer the content of the image.

Example

[Example 1]

<Question>

Requirements

- The description should be brief but precise.
- If both the answer and the response are long, focus on describing the one or two most significant differences.
- Do not provide any analysis or explanation; only describe the image.
- A common approach is to describe what is present in the image and what is missing.

<Question>

{question}

<Answer>

{answer}

<Response>

{response}

<Output Description>

Example 생략

4. Generate
LLM + Diffusion



Calculate standard deviation: σ_r

$$\tau = \max(\delta, 2\sigma_r)$$

Select all non-overlapping pairs:

$$r^+ - r^- > \tau, r^+ > \tau_{pos}, r^- < \tau_{neg}$$

Exist?

Yes

No

$$r_{worst} < \tau_{neg}?$$

Yes

The duck's beak is white.

The duck's beak is orange.

The duck in the

[Worst response]

02. Method

Sample

Evaluate

Filter

Generate

Online Training

5. Dynamic Inference and Experience Buffer & Preference Optimization

Experience Buffer

- Batch 단위 학습 안정화 및 Online Sampling의 유연성 유지를 위해 Contrastive Training sample을 저장하는 Experience Buffer 도입

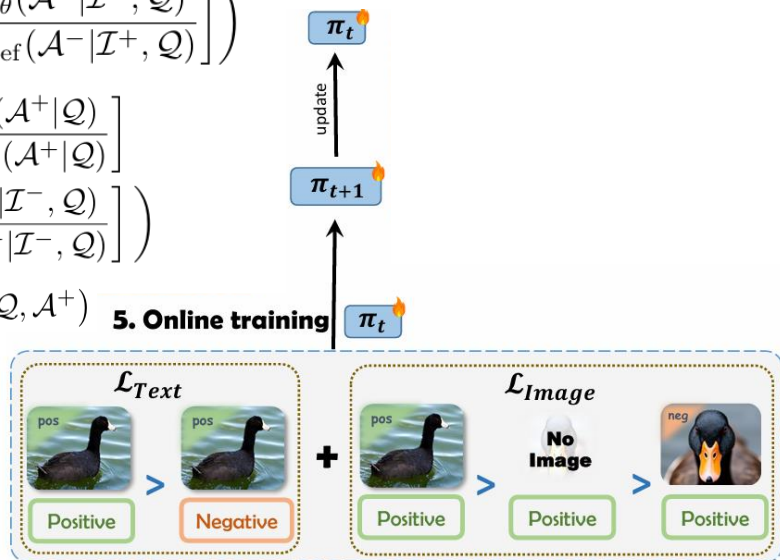
Image- and Response-Side Preference Optimization

$$\mathcal{L}_{\text{Text}}(\mathcal{A}^+, \mathcal{A}^-; \mathcal{I}^+, \mathcal{Q}) = \log \sigma \left(\beta \cdot \left[\log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{I}^+, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{I}^+, \mathcal{Q})} - \log \frac{\pi_{\theta}(\mathcal{A}^- | \mathcal{I}^+, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^- | \mathcal{I}^+, \mathcal{Q})} \right] \right)$$

$$\mathcal{L}_{\text{Image}}(\mathcal{I}^+, \mathcal{I}^-; \mathcal{Q}, \mathcal{A}^+) = -\log \sigma \left(\beta_1 \cdot \left[\log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{I}^+, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{I}^+, \mathcal{Q})} - \log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{Q})} \right] + \beta_2 \cdot \left[\log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{Q})} - \log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{I}^-, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{I}^-, \mathcal{Q})} \right] \right)$$

$$\mathcal{L}_{\text{OViP}}(\mathcal{Q}, \mathcal{I}^+, \mathcal{I}^-, \mathcal{A}^+, \mathcal{A}^-) = \mathcal{L}_{\text{Text}}(\mathcal{A}^+, \mathcal{A}^-; \mathcal{I}, \mathcal{Q}) + \mathcal{L}_{\text{Image}}(\mathcal{I}^+, \mathcal{I}^-; \mathcal{Q}, \mathcal{A}^+)$$

5. Online training



02. Method

Sample

Evaluate

Filter

Generate

Online
Training

5. Dynamic Inference and Experience Buffer & Preference Optimization

» Symmetrical Visual Contrastive Optimization

- Contrastive Learning 방식을 개선하여 Symmetrical 하도록 Contrastive Optimization을 수행하는 방식
 - A 이미지와 A 텍스트는 Positive + B 이미지와 A 텍스트는 Negative + B 이미지와 B 텍스트는 Positive 하다
- 이미지 A는 항상 Positive하다 와 같은 Shortcut 방지, Image-text 간 Context 정확하게 파악하도록 학습
- 이미지의 미묘한 시각적 차이를 구분하도록 학습됨

OURS



S-VCO: “preference”/contrasts over image-text alignment pair

03

Experiments

03. Experiments

1. Setup

» Implementation Details

- **LLaVA-1.5-7B-hf / 13B**
 - Visual Encoder : CLIP ViT-L-336px
 - LLM : Vicuna-6b/13b

» Training

- **OPA-DPO 논문에서 사용한 Dataset 사용**
 - RLAI-F-V 논문에서 사용한 데이터셋의 Subset
 - MSCOCO, ShareGPT-4V, MovieNet, Google Landmark v2, VQAv2에서 수집한 데이터셋
- **7B/13B 모델 모두 1 Epoch 학습, LoRA(Rank:256, alpha:512) 적용**

» Baselines

- **Methods : SFT, DP, mDPO, GPRO, HA-DPO, HALVA, RLAI-F-V, OPADPO**

03. Experiments

2. Evaluation Metrics



Hallucination Related Evaluation

- **MMHal : 8 Categories, 12 Topic에서 Question-Answering을 평가하는 Metric**
 - gpt-4o-2024-05-13을 사용
- **AMB_{gen} : Image description task를 평가/비교 (judging-model-free benchmark)**
 - Chair : Object-level의 hallucination rate 평가 (Precision)
 - Cover : Object recall 평가
 - 두 Metric이 trade-off이므로 본 논문에서는 Chair와 Cover 간 F1 Score를 제시
- **ObjectHal : Object-level의 완성도와 hallucination rate를 평가.**
 - 본 논문에서는 F1-Score 사용 제시
- **LV : Visual Understanding, Reasoning, Instruction following의 능력 측정**
 - gpt-4o-2024-05-13을 사용
- **$AMBER_{dis}$: Yes/No Question을 기반 평가**
 - 본 논문에서는 F1-Score 사용 제시

03. Experiments

2. Evaluation Metrics

» Hallucination Related Evaluation

- HRI(Hallucination Reduction Index)
 - Baseline 대비 증감폭을 Rescaling하고, 0~2범위로 근사화하여 최종 HRI Score 제시

$$\text{HRI} = 2 \times \sum_{i=1}^5 \frac{a_i - a_i^{\text{base}}}{a_i^{\text{ref}} - a_i^{\text{base}}}$$

For 7B model, we set the reference performances as **OVIP**_{2ep}, so it comes:

$$\text{HRI} = 2 \times \left(\frac{a_1 - 65.01}{67.12 - 65.01} + \frac{a_2 - 1.90}{2.65 - 1.90} + \frac{a_3 - 72.40}{74.18 - 72.40} + \frac{a_4 - 57.20}{60.90 - 57.20} + \frac{a_5 - 85.5}{87.4 - 85.5} \right)$$

» Out-of-Domain General Capability Evaluation

- Non-hallucination Benchmark에서 평가 수행 및 여러 지표를 통합적으로 계산한 Accuracy Difference 제시
 - realworldQA, TextVQA, CVBench, MMStar

03. Experiments

3. Main Result

		AMB _{gen}		MMHal	ObjectHal		LV	AMB _{dis}	HRI	General	
	Chair↓	Cover↑	F1↑	Score↑	Chair _r ↓	F1↑	Score↑	F1↑		Acc _{Dif}	
	GPT4-V [†]	4.6	67.1	78.8	3.49 [‡]	13.6	-	95.3	87.4	-	-
LLaVA-1.5-7B	Baseline	7.1	50.0	65.01	1.90	51.38	72.40	57.20	85.5	-	-
	HA-DPO* 31	5.6	49.4	64.86	1.95	37.15	73.81	57.30	85.4	1.52	-11.59
	HALVA* 18	5.7	52.9	67.78	2.12	43.40	76.01	58.60	86.5	<u>9.08</u>	-7.36
	RLAIF-V* [#] 16	3.1	49.8	65.79	2.54	9.35	69.78	58.90	86.4	1.37	-6.74
	OPA-DPO* [#] 28	2.4	45.2	61.79	2.78	6.37	63.26	64.80	86.7	-5.60	-11.82
	SFT	3.5	50.6	66.39	2.52	20.60	70.30	52.20	86.1	-1.47	-8.07
	DPO	3.7	48.9	64.86	2.35	26.60	71.95	56.70	86.8	1.65	-3.86
	mDPO	3.4	48.6	64.67	2.55	25.45	73.92	55.80	86.1	2.99	-3.05
	DPO _{iterative}	3.9	48.7	64.64	2.32	27.11	72.33	56.40	86.5	1.31	-2.98
	GRPO _{2ep}	4.8	51.2	66.59	2.45	34.98	73.83	58.70	86.8	6.75	-3.83
	OViP	4.0	51.1	66.70	2.52	33.22	73.50	63.10	87.3	<u>9.58</u>	+0.88
	OViP_{2ep}	4.0	51.6	67.12	2.65	29.54	74.18	60.90	87.4	<u>10.00</u>	-1.01
LLaVA-1.5-13B	Baseline	6.5	51.0	65.99	2.24	46.18	76.73	62.60	89.1	-	-
	HALVA* 18	6.0	52.2	67.12	2.45	35.07	77.75	61.70	90.0	4.22	-5.45
	OPA-DPO* [#] 28	2.8	47.8	64.08	2.88	5.88	64.46	64.70	89.3	-7.05	-15.25
	SFT	4.5	50.0	65.64	2.38	31.21	75.81	64.00	89.9	1.79	-1.24
	DPO	3.6	50.6	66.37	2.53	25.00	75.00	65.30	89.6	2.42	+0.12
	mDPO	3.9	50.1	65.86	2.51	21.79	75.35	64.50	89.5	1.78	-1.12
	GRPO _{2ep}	3.8	52.4	67.84	2.38	23.76	75.55	66.70	90.4	4.96	-1.48
	OViP	4.4	53.1	68.28	2.58	36.30	76.52	64.60	89.7	<u>5.25</u>	+0.85
	OViP_{2ep}	3.6	53.7	68.98	2.57	28.62	76.75	67.90	90.2	<u>8.02</u>	+2.02

03. Experiments

4. Ablation Study

» The Impact of Loss Functions

$$\mathcal{L}_{\text{Image}}^{\text{base}}(\mathcal{I}^+, \mathcal{I}^-; \mathcal{Q}, \mathcal{A}^+) = \log \sigma \left(\beta \cdot \left[\log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{I}^+, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{I}^+, \mathcal{Q})} - \log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{I}^-, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{I}^-, \mathcal{Q})} \right] \right)$$

$$\begin{aligned} \mathcal{L}_{\text{Image-Sym}}(\mathcal{I}^+, \mathcal{I}^-, \mathcal{A}^+, \mathcal{A}^-; \mathcal{Q}) &= \mathcal{L}_{\text{Image}}(\mathcal{I}^+, \mathcal{I}^-; \mathcal{Q}, \mathcal{A}^+) + \mathcal{L}_{\text{Image}}(\mathcal{I}^-, \mathcal{I}^+; \mathcal{Q}, \mathcal{A}^-) \\ &= -\log \sigma \left(\beta_1 \cdot \left[\log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{I}^+, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{I}^+, \mathcal{Q})} - \log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{Q})} \right] \right. \\ &\quad \left. + \beta_2 \cdot \left[\log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{Q})} - \log \frac{\pi_{\theta}(\mathcal{A}^+ | \mathcal{I}^-, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^+ | \mathcal{I}^-, \mathcal{Q})} \right] \right) \\ &\quad -\log \sigma \left(\beta_1 \cdot \left[\log \frac{\pi_{\theta}(\mathcal{A}^- | \mathcal{I}^-, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^- | \mathcal{I}^-, \mathcal{Q})} - \log \frac{\pi_{\theta}(\mathcal{A}^- | \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^- | \mathcal{Q})} \right] \right. \\ &\quad \left. + \beta_2 \cdot \left[\log \frac{\pi_{\theta}(\mathcal{A}^- | \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^- | \mathcal{Q})} - \log \frac{\pi_{\theta}(\mathcal{A}^- | \mathcal{I}^+, \mathcal{Q})}{\pi_{\text{ref}}(\mathcal{A}^- | \mathcal{I}^+, \mathcal{Q})} \right] \right) \end{aligned}$$

03. Experiments

4. Ablation Study

» The Impact of Loss Functions

Loss Functions	<i>Training from scratch</i>				<i>Iterative training</i>			
	AMB _{gen}	MMHal	AMB _{dis}	HRI	ObjectHal	MMHal	AMB _{dis}	HRI
OViP	65.38	2.35	86.6	4.32	74.14	2.70	85.7	7.94
– $\mathcal{L}_{\text{Text}}$	66.61	2.12	85.4	4.23	74.01	2.60	86.1	7.71
– $\mathcal{L}_{\text{Image}}$	63.58	2.06	85.9	-2.29	72.63	2.40	86.6	4.56
$\mathcal{L}_{\text{Text}} + \mathcal{L}_{\text{Image}}^{\text{base}}$	65.10	2.17	86.2	4.08	73.45	2.58	86.3	7.50
$\mathcal{L}_{\text{Image-Sym}}$	65.54	2.09	85.6	-0.32	73.86	2.41	86.3	6.57

04

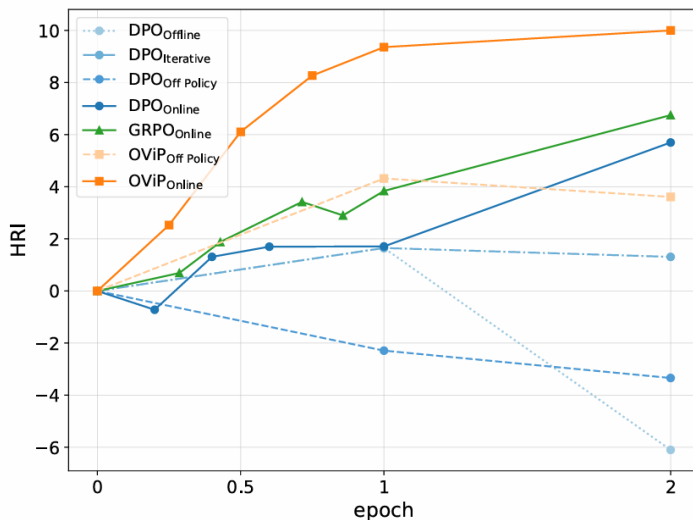
Conclusion

04. Conclusion



Contribution

- ▶ 시각적인 Hallucination 현상에 대한 고찰
- ▶ 다양한 실험을 통한 Hallucination 극복과 General Performance 성능 저하 여부 확인
- ▶ 향후 장기적인 Online 학습 전략 안정성 및 일반화 성능 탐구 방향 제시

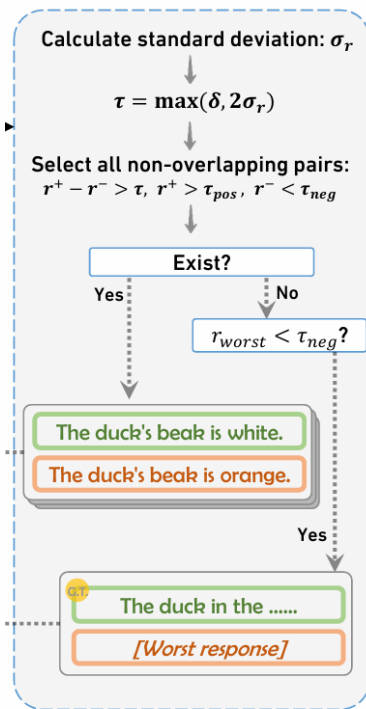


04. Conclusion



▶ Sample 과정에서 Negative가 Threshold 이하인 것이 없는 경우?

Discussion



while $\exists (\mathcal{A}^+, \mathcal{A}^-)$ satisfying:

$|r^+ - r^-| > \max(\delta, 2\sigma_r), r^+ > \tau_{pos}, r^- < \tau_{neg}$ **do**

Add $(\mathcal{A}^+, \mathcal{A}^-)$ to $\mathcal{T}$ and remove from candidate pool

end while

if $\mathcal{T} = \emptyset$ and $\min_i r^i < \tau_{neg}$ **then**

Let $\mathcal{A}^-$ be the lowest-scoring response

Add $(\mathcal{A}^*, \mathcal{A}^-)$ to $\mathcal{T}$

endif

for each $(\mathcal{A}^+, \mathcal{A}^-) \in \mathcal{T}$ **do**

Generate prompt: $\mathcal{T}^- = G_{\text{diff}}(\mathcal{A}^+, \mathcal{A}^-)$

Synthesize image: $\mathcal{I}^- = \text{diff}(\mathcal{T}^-)$

Add $(\mathcal{I}^+, \mathcal{I}^-, \mathcal{Q}, \mathcal{A}^+, \mathcal{A}^-)$ to buffer $\mathcal{B}$

end for



전부 다 Negative Sample인 경우,
Hallucination 현상이 거의 없는 것으로 간주하여 학습 Pass ?

04. Conclusion



Discussion

▶ 이미지 생성 AI 모델의 Hallucination에 대한 대응 방법?

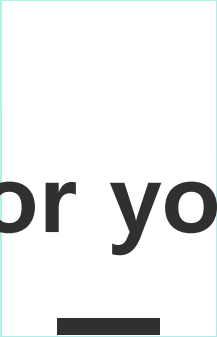
Prompt for image distortion-Part 1

Task

Given an unknown image-related question, a correct answer, and an inaccurate response, carefully analyze the differences between the response and the answer. Then, provide a brief description of the image so that it aligns with the correct answer and differs from the incorrect responses. In other words, infer the content of the image.

- 최대한 Prompt 충실도가 높도록 만들도록 하여 Hallucination이 존재하지 않도록 함
- Negative Sample은 설명을 제외한 부분에서 자유도가 있더라도 상관 없지 않을까 ?

Q & A



Thank you for your attention